

Bluerov硬件升级&环境配置&使用说明

1.配置小主机Intel NUC

装ubuntu20.04系统，安装ROS（[安装教程](#)）

2.安装声呐驱动&声呐使用

1. 首先安装`oculus_driver`：，它的用处是在没有网络连接的情况下也可以使用`oculus_sonar`。按照链接中的readme安装即可

```
fyx@fyx-virtual-machine: ~  
fyx@fyx-virtual-machine:~$ sudo apt-get install -y libboost-system-dev libboost-thread-dev  
[sudo] fyx 的密码:  
正在读取软件包列表... 完成  
正在分析软件包的依赖关系树  
正在读取状态信息... 完成  
libboost-system-dev 已经是最新版 (1.71.0.0ubuntu2)。  
libboost-thread-dev 已经是最新版 (1.71.0.0ubuntu2)。  
下列软件包是自动安装的并且现在不需要了:  
  linux-headers-5.15.0-50-generic linux-hwe-5.15-headers-5.15.0-50  
  linux-image-5.15.0-50-generic linux-modules-5.15.0-50-generic  
  linux-modules-extra-5.15.0-50-generic  
使用'sudo apt autoremove'来卸载它(它们)。  
升级了 0 个软件包，新安装了 0 个软件包，要卸载 0 个软件包，有 433 个软件包未被升级。  
fyx@fyx-virtual-machine:~$ git clone https://gitee.com/fengyunxuan123/oculus_driver  
正克隆到 'oculus_driver'...  
Username for 'https://gitee.com': fengyunxuan123  
Password for 'https://fengyunxuan123@gitee.com':  
remote: Enumerating objects: 1250, done.  
remote: Counting objects: 100% (1250/1250), done.  
remote: Compressing objects: 100% (382/382), done.  
remote: Total 1250 (delta 750), reused 1250 (delta 750), pack-reused 0  
接收对象中: 100% (1250/1250), 290.42 KiB | 411.00 KiB/s, 完成。  
处理 delta 中: 100% (750/750), 完成。  
fyx@fyx-virtual-machine:~$ cd oculus_driver/  
fyx@fyx-virtual-machine:~/oculus_driver$ mkdir build && cd build  
fyx@fyx-virtual-machine:~/oculus_driver/build$ cmake -DCMAKE_BUILD_TYPE=Release  
-DCMAKE_INSTALL_PREFIX=/home/fyx/oculus_driver ..  
-- The C compiler identification is GNU 9.4.0  
-- The CXX compiler identification is GNU 9.4.0  
-- Check for working C compiler: /usr/bin/cc  
-- Check for working C compiler: /usr/bin/cc -- works  
-- Detecting C compiler ABI info  
-- Detecting C compiler ABI info - done  
-- Detecting C compile features  
-- Detecting C compile features - done
```

```
-- Check for working CXX compiler: /usr/bin/c++
-- Check for working CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Found Boost: /usr/lib/x86_64-linux-gnu/cmake/Boost-1.71.0/BoostConfig.cmake
(found version "1.71.0") found components: system thread
-- Configuring done
-- Generating done
-- Build files have been written to: /home/fyx/oculus_driver/build
fyx@fyx-virtual-machine:~/oculus_driver/build$ make -j4 install
Scanning dependencies of target oculus_driver
[ 42%] Building CXX object CMakeFiles/oculus_driver.dir/src/SonarClient.cpp.o
[ 42%] Building CXX object CMakeFiles/oculus_driver.dir/src/StatusListener.cpp.o
[ 57%] Building CXX object CMakeFiles/oculus_driver.dir/src/SonarDriver.cpp.o
[ 57%] Building CXX object CMakeFiles/oculus_driver.dir/src/print_utils.cpp.o
[ 71%] Building CXX object CMakeFiles/oculus_driver.dir/src/AsyncService.cpp.o
[ 85%] Building CXX object CMakeFiles/oculus_driver.dir/src/Recorder.cpp.o
[100%] Linking CXX shared library liboculus_driver.so
[100%] Built target oculus_driver
Install the project...
-- Install configuration: "Release"
-- Installing: /home/fyx/oculus_driver/lib/liboculus_driver.so
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverTar
gets.cmake
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverTar
gets-release.cmake
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverCon
fig.cmake
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverCon
figVersion.cmake
```

在./bashrc文件中添加：

```
2 export CMAKE_PREFIX_PATH=/home/fyx/oculus_driver:$CMAKE_PREFIX_PATH
```

```
fyx@NUC11PAH15:~$ source ~/.bashrc
fyx@NUC11PAH15:~$ echo $CMAKE_PREFIX_PATH
/home/fyx/oculus_driver:/opt/ros/noetic:/home/fyx/oculus_driver
```

图中代码根据安装位置适当更换

2.安装oculus_ros功能包

oculus_sonar功能包

创建好一个工作空间，把上面链接里面的文件下载到src文件夹里面解压好，返回工作空间catkin_make

运行launch文件的代码：`roslaunch oculus_sonar sonar_oculus_ros.launch`

launch文件里面有一个路径更改：

```

<!-- 打开rviz -->
<node pkg="rviz" type="rviz" name="rviz" args="-d ~/oculus_ros/src/sonar_oculus/cfg/-
sonar_rviz.rviz" required="true" />

```

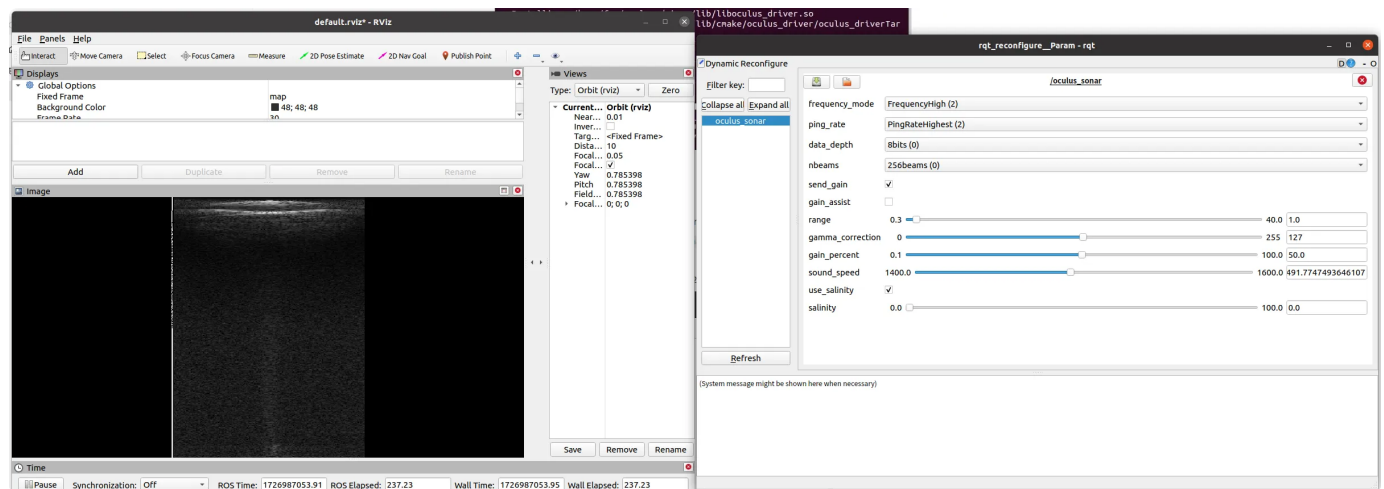
oculus_viewer.py允许文件作为程序执行打钩

使用：

插上网线后，有线连接的IPV4地址改为：169.254.70.29（也可能是169.254.179.91）



调整参数后就可以在rviz中看到图像：



3.安装XSENS MTI300 IMU驱动

XSENS驱动软件包下载，解压

打开终端, `sudo apt-get install sharutils`

~/下载/MT_Software_Suite_linux-x64_2020.5路径下, `sudo ./mtsdk_linux-x64_2020.5.sh`

创建好一个工作空间imu_ws, 将/usr/local/xsens文件夹中的xsens_ros_mti_driver文件夹复制到工作空间src下

~/imu_ws/src/xsens_ros_mti_driver/lib/xspublic路径下, `make`

返回工作空间目录, `catkin_make`

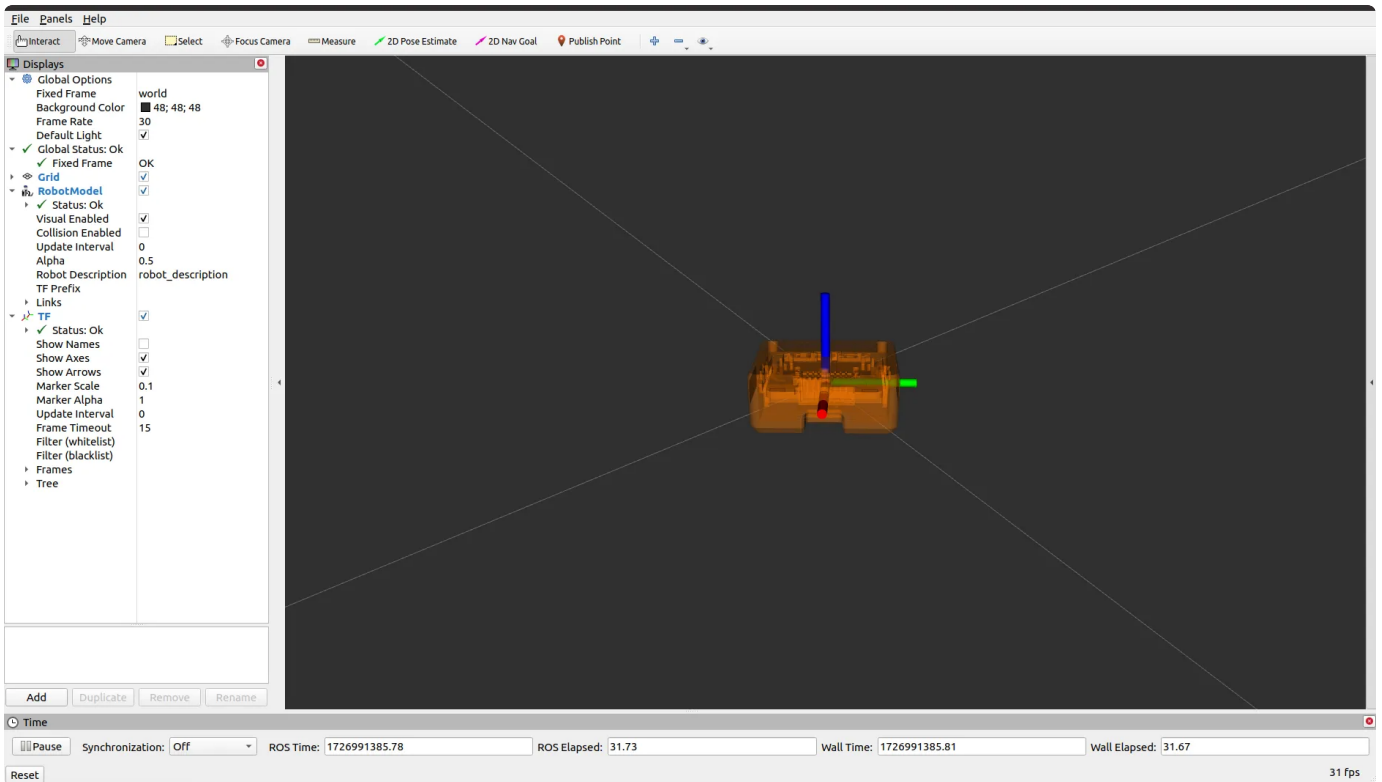
```
hit@hit-NUC8i5BEHS:~/imu_ws$ catkin_make
Base path: /home/hit/imu_ws
Source space: /home/hit/imu_ws/src
Build space: /home/hit/imu_ws/build
Devel space: /home/hit/imu_ws/devel
Install space: /home/hit/imu_ws/install
####
#### Running command: "make cmake_check_build_system" in "/home/hit/imu_ws/build"
####
####
#### Running command: "make -j8 -l8" in "/home/hit/imu_ws/build"
####
[ 25%] Linking CXX executable /home/hit/imu_ws/devel/lib/xsens_mti_driver/xsens_mti_node
[100%] Built target xsens_mti_node
```

imu连接设备, 查看串口编号, 并给权限:

```
hit@hit-NUC8i5BEHS: ~
hit@hit-NUC8i5BEHS:~$ ls /dev/ttyUSB*
/dev/ttyUSB0
hit@hit-NUC8i5BEHS:~$ sudo chmod 777 /dev/ttyUSB0
[sudo] hit 的密码:
hit@hit-NUC8i5BEHS:~$
```

`roslaunch xsens_mti_driver display.launch` 就可以启动设备


```
hit@hit-NUC8i5BEHS:~$ rostopic list
/clicked_point
/filter/free_acceleration
/filter/positionlla
/filter/quaternion
/filter/twist
/filter/velocity
/gnss
/imu/acceleration
/imu/angular_velocity
/imu/data
/imu/dq
/imu/dv
/imu/mag
/imu/time_ref
/initialpose
/move_base_simple/goal
/pressure
/rosout
/rosout_agg
/temperature
/tf
/tf_static
hit@hit-NUC8i5BEHS:~$
```

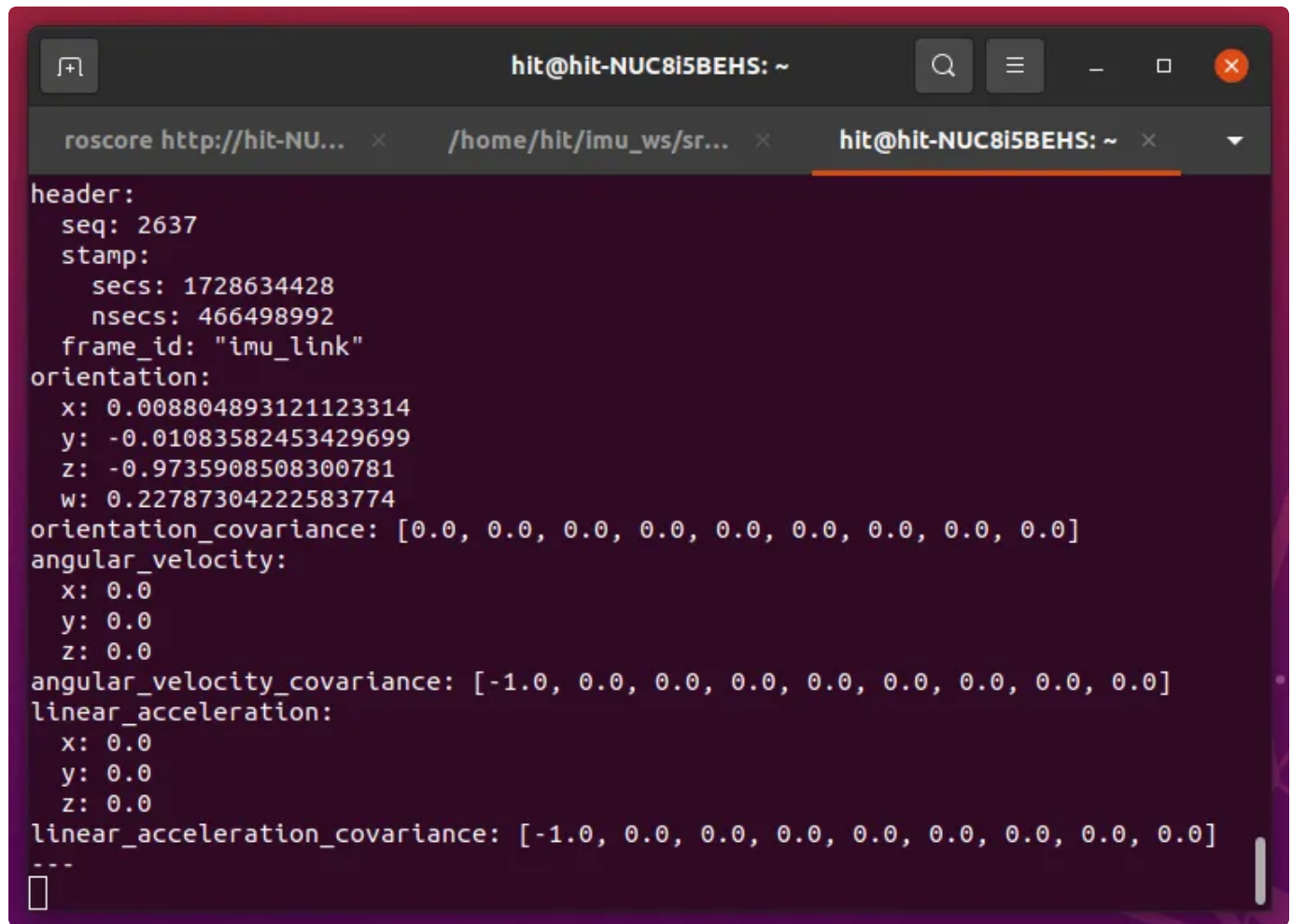


现在每次开机插入设备都需要重新给设备权限, `sudo chmod 777 /dev/ttyUSB0`

解决方法:

```
sudo usermod -aG dialout $USER (用户名) , 重启电脑
```

ROS输出没有角速度和线加速度信息：

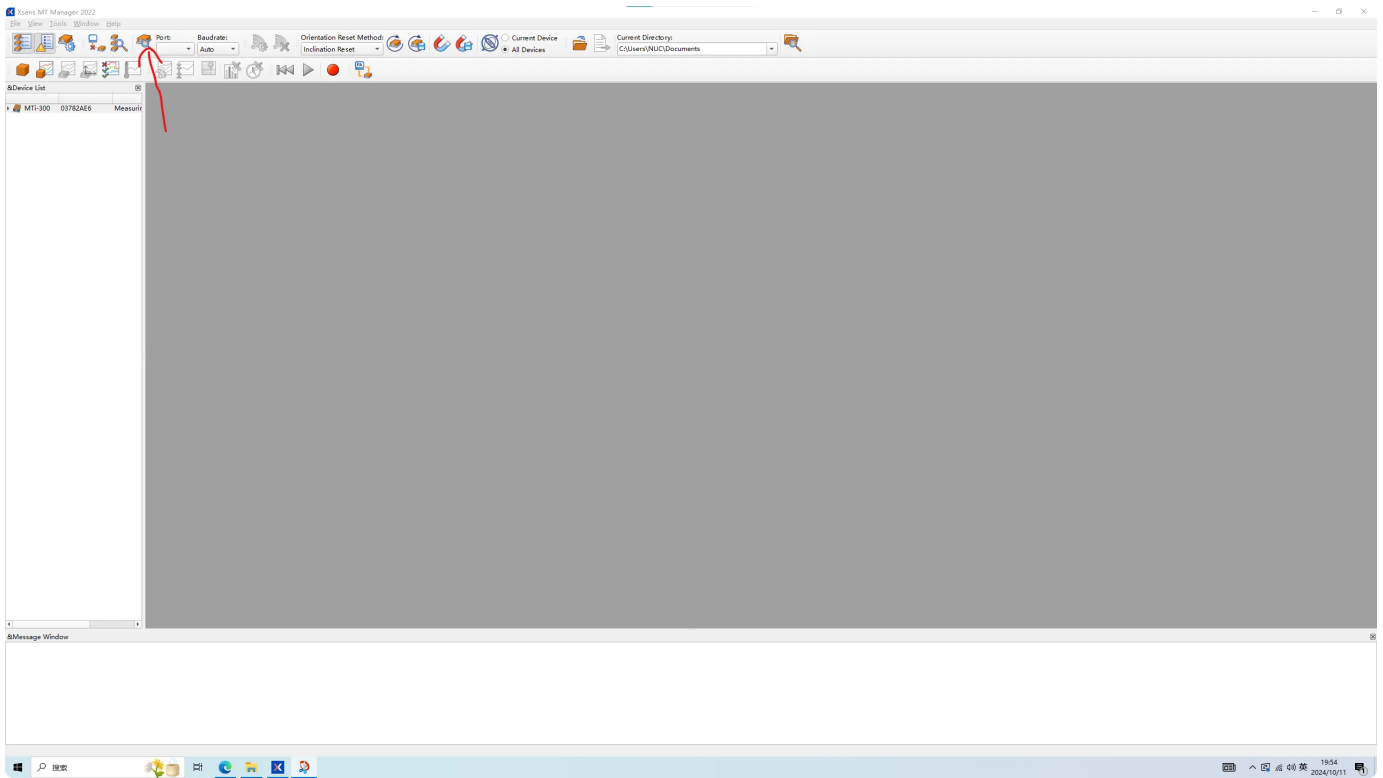


```
header:
  seq: 2637
  stamp:
    secs: 1728634428
    nsecs: 466498992
  frame_id: "imu_link"
orientation:
  x: 0.008804893121123314
  y: -0.01083582453429699
  z: -0.9735908508300781
  w: 0.22787304222583774
orientation_covariance: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
angular_velocity:
  x: 0.0
  y: 0.0
  z: 0.0
angular_velocity_covariance: [-1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
linear_acceleration:
  x: 0.0
  y: 0.0
  z: 0.0
linear_acceleration_covariance: [-1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
---
```

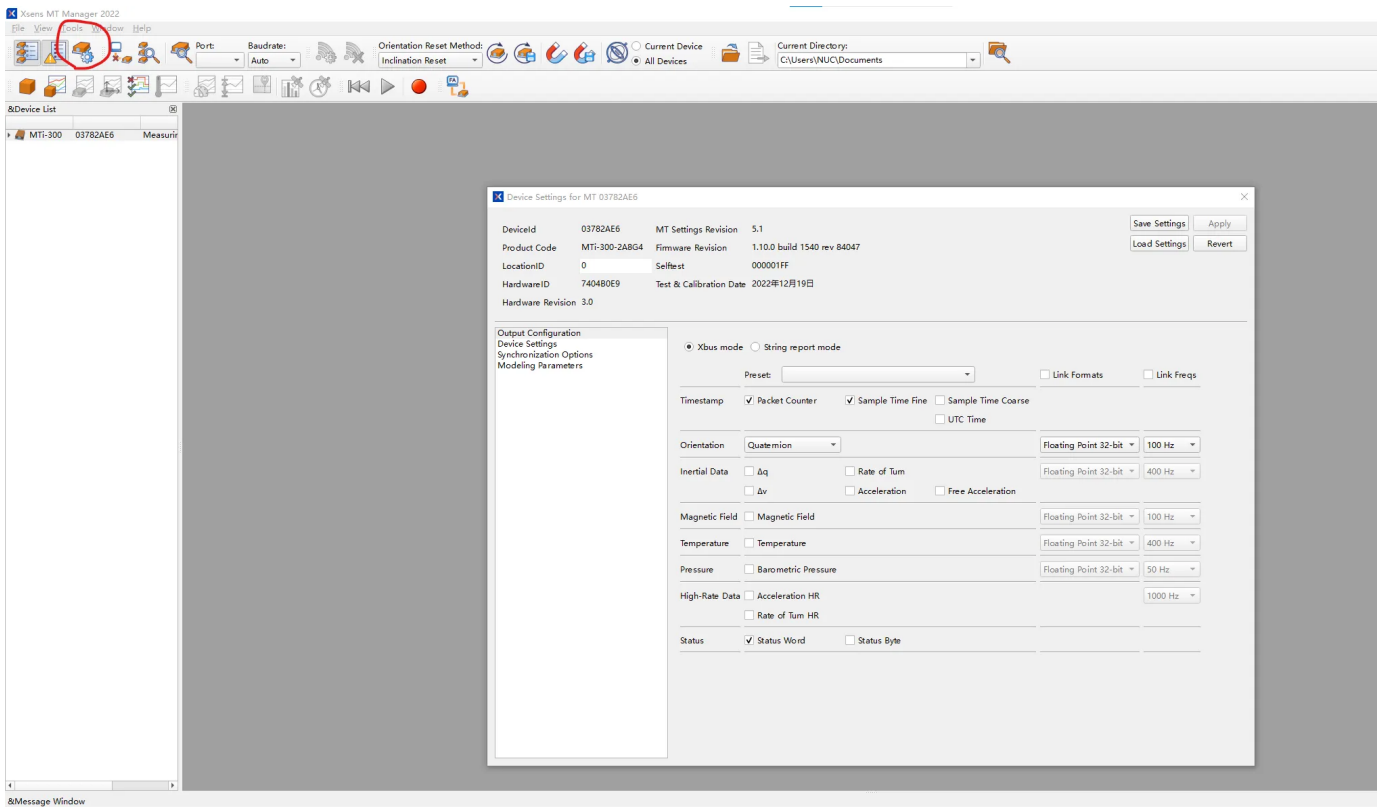
需要使用MT Manager来更改输出配置

[MT Manager下载网站](#)，下载windows版本就可以

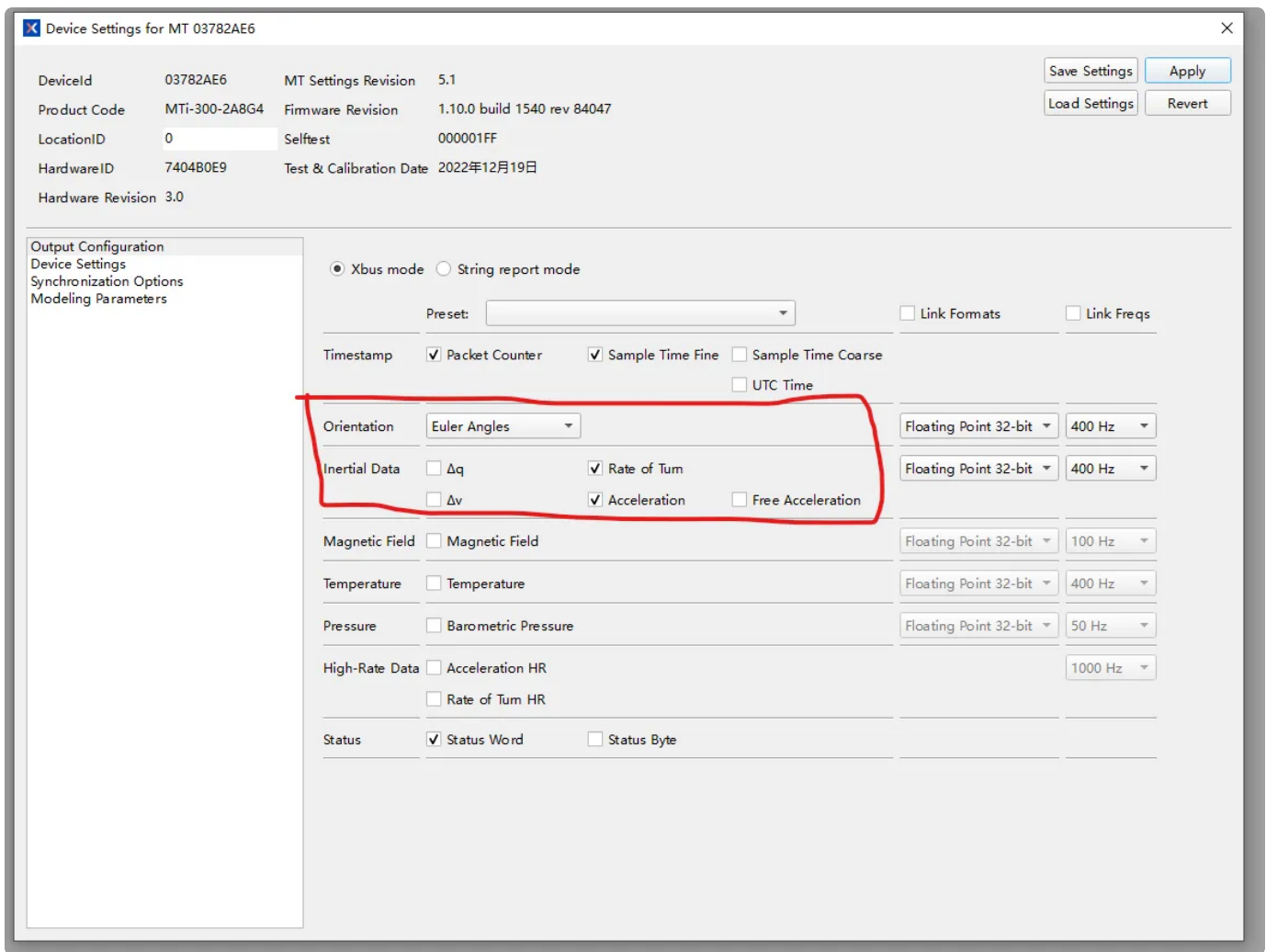
打开界面，连接IMU，点击扫描端口按钮，扫描设备



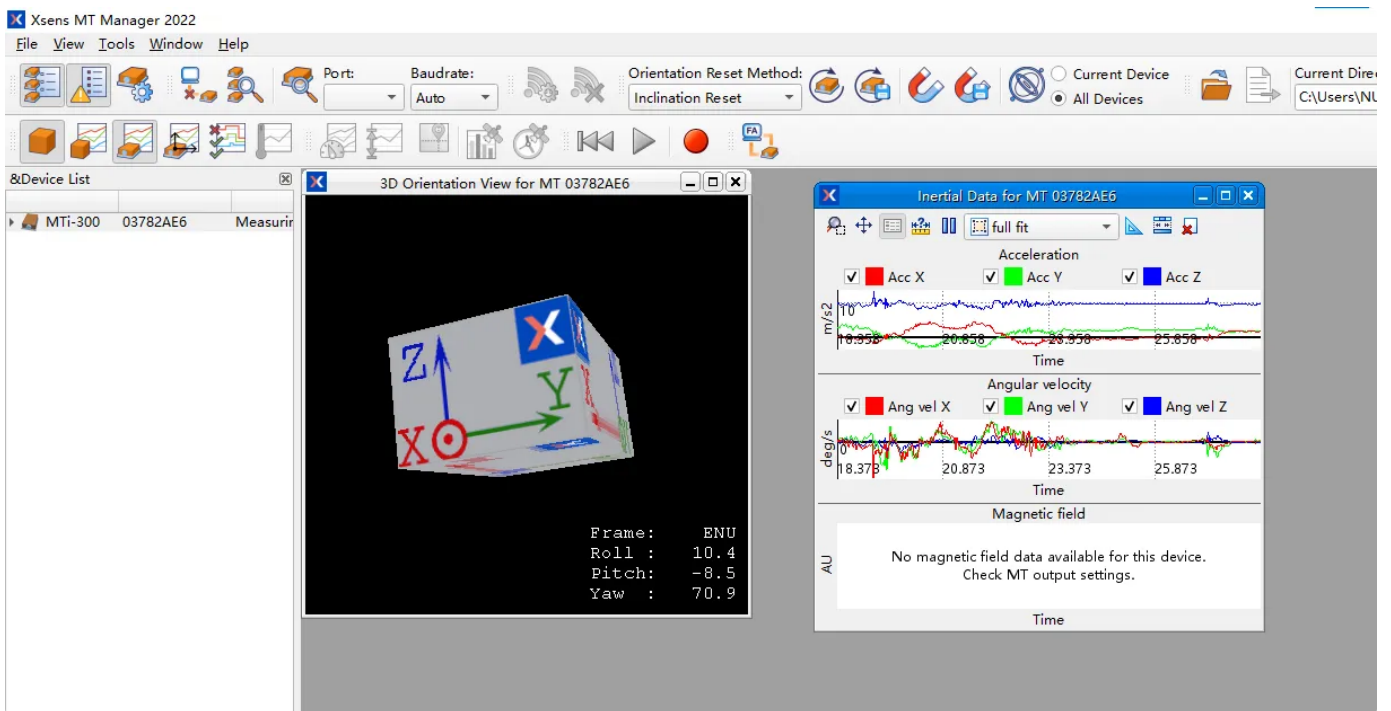
左边选中设备，点击配置参数按钮



选择角速度和加速度，右上角应用



可以看见姿态显示与角速度、加速度输出



回到ubuntu ros中也有输出了:

```
hit@hit-NUC8i5BEHS: ~  
roscore http://hit-NU... x /home/hit/imu_ws/src... x hit@hit-NUC8i5BEHS: ~ x  
header:  
  seq: 27558  
  stamp:  
    secs: 1728648417  
    nsecs: 455058495  
  frame_id: "imu_link"  
orientation:  
  x: -0.001326167962815489  
  y: -0.0033514436568834786  
  z: 0.5596528903841431  
  w: 0.8287192838281388  
orientation_covariance: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]  
angular_velocity:  
  x: -0.0014200806617736816  
  y: -0.0012293459149077537  
  z: 0.0007033348083496094  
angular_velocity_covariance: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]  
linear_acceleration:  
  x: 0.03066079877316952  
  y: -0.03495711088180542  
  z: 9.773182868957521  
linear_acceleration_covariance: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]  
---
```

4.安装BlueRov驱动与ROS功能包

[bluerov_ros_playground](#)

创建好一个工作空间bluerov_ws，bluerov_ws/src文件夹下，克隆上面的功能包，返回工作空间
catkin_make

安装[QGC](#)

USB以太网IPV4设置：192.168.2.1



运行：`roslaunch bluerov_ros_playground user_mav.launch`

出现以下错误是因为没有安装mavros包，按照[mavros安装](#)即可

```
hit@hit-NUC8i5BEHS:~$ roslaunch bluerov_ros_playground user_mav.launch
... logging to /home/hit/.ros/log/4107bcf4-7e64-11ef-9c9e-33ec38f11f36/roslaunch
-hit-NUC8i5BEHS-11541.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

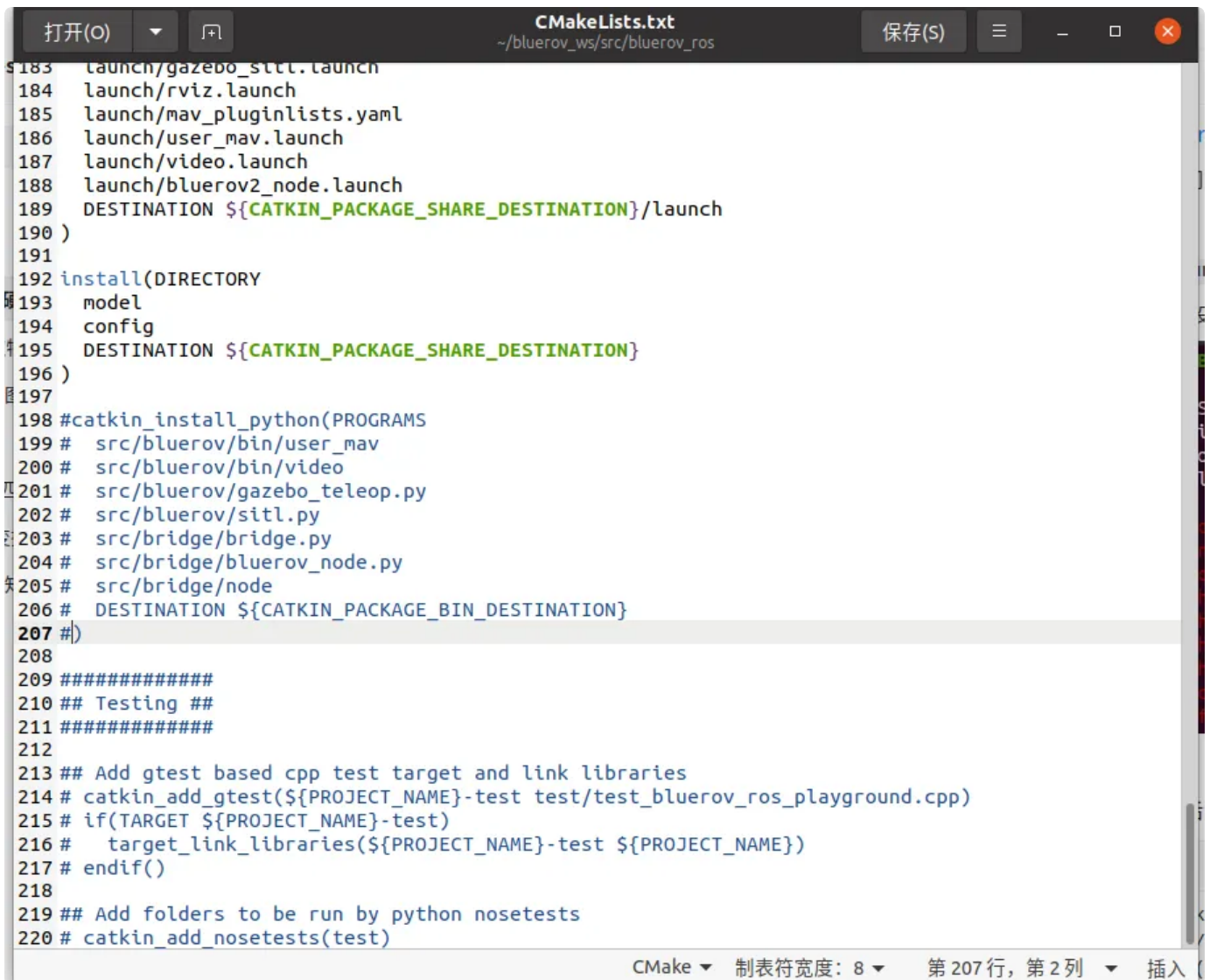
Resource not found: The following package was not found in <arg name="config_yam
l" value="$(find mavros)/launch/apm_config.yaml"/>: mavros
ROS path [0]=/opt/ros/noetic/share/ros
ROS path [1]=/home/hit/dvl_ws/src
ROS path [2]=/home/hit/bluerov_ws/src
ROS path [3]=/home/hit/imu_ws/src
ROS path [4]=/home/hit/oculus_ros/src
ROS path [5]=/opt/ros/noetic/share
The traceback for the exception was written to the log file
```

上面安装的包运行后会报错：

```
1 Traceback (most recent call last):
2   File "/home/hit/bluerov_ws/devel/lib/bluerov_ros_playground/node", line
  15, in <module>
3     exec(compile(fh.read(), python_script, 'exec'), context)
4   File "/home/hit/bluerov_ws/src/bluerov_ros/src/bridge/node", line 4, in
  <module>
5     import bluerov_node as BR
6   File "/home/hit/bluerov_ws/devel/lib/bluerov_ros_playground/bluerov_nod
  e.py", line 15, in <module>
7     exec(compile(fh.read(), python_script, 'exec'), context)
8   File "/home/hit/bluerov_ws/src/bluerov_ros/src/bridge/bluerov_node.py",
  line 12, in <module>
9     from bridge import Bridge
10 ImportError: cannot import name 'Bridge' from 'bridge' (/home/hit/bluerov_
  ws/devel/lib/bluerov_ros_playground/bridge.py)
```

解决办法:

/home/hit/bluerov_ws/src/bluerov_ros/CMakeList.txt文件198-207行注释掉:



```
183 launch/gazebo_sitl.launch
184 launch/rviz.launch
185 launch/mav_pluginlists.yaml
186 launch/user_mav.launch
187 launch/video.launch
188 launch/bluerov2_node.launch
189 DESTINATION ${CATKIN_PACKAGE_SHARE_DESTINATION}/launch
190 )
191
192 install(DIRECTORY
193   model
194   config
195   DESTINATION ${CATKIN_PACKAGE_SHARE_DESTINATION}
196 )
197
198 #catkin_install_python(PROGRAMS
199 #  src/bluerov/bin/user_mav
200 #  src/bluerov/bin/video
201 #  src/bluerov/gazebo_teleop.py
202 #  src/bluerov/sitl.py
203 #  src/bridge/bridge.py
204 #  src/bridge/bluerov_node.py
205 #  src/bridge/node
206 #  DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
207 #)
208
209 #####
210 ## Testing ##
211 #####
212
213 ## Add gtest based cpp test target and link libraries
214 # catkin_add_gtest(${PROJECT_NAME}-test test/test_bluerov_ros_playground.cpp)
215 # if(TARGET ${PROJECT_NAME}-test)
216 #   target_link_libraries(${PROJECT_NAME}-test ${PROJECT_NAME})
217 # endif()
218
219 ## Add folders to be run by python nosetests
220 # catkin_add_nosetests(test)
```

另外，将所有.py文件的第一行由 `#!/usr/bin/env python` 改为 `#!/usr/bin/env python3`

更改后，运行 `roslaunch bluerov_ros_playground user_mav.launch` 可以正常运行，输出数据并弹出视觉摄像头窗口：

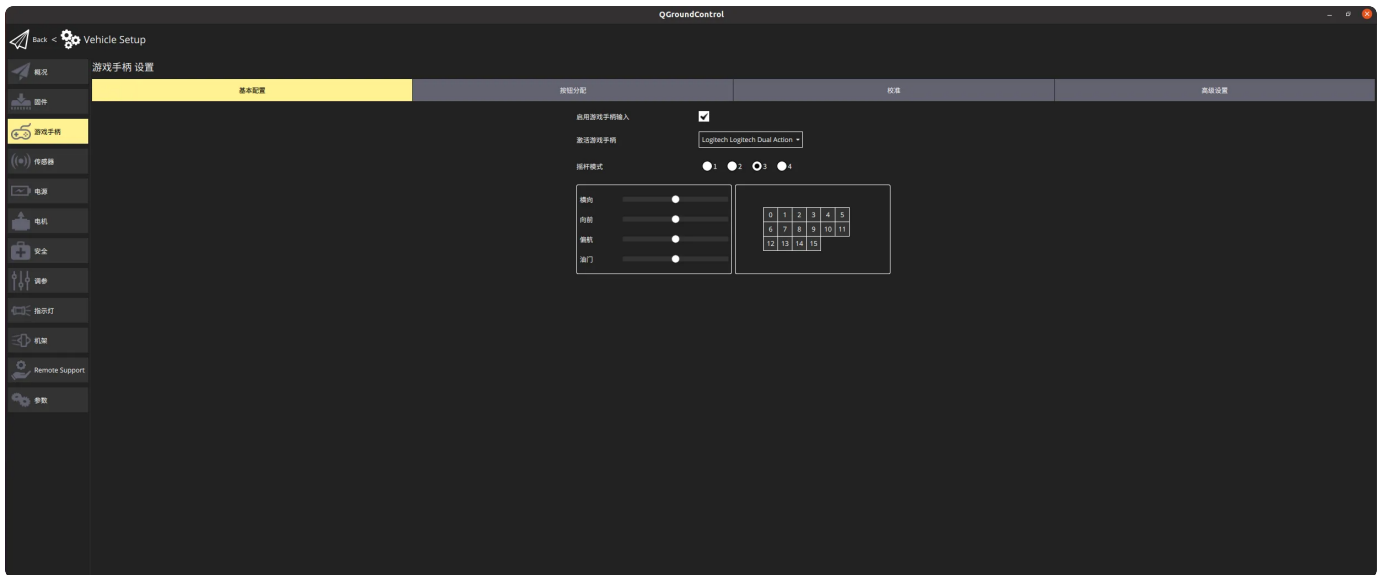


`rostopic echo /mavros/imu/data` 输出（机器人自带imu数据）：

```
---
header:
  seq: 1811
  stamp:
    secs: 1727621179
    nsecs: 25567073
  frame_id: "base_link"
orientation:
  x: -0.05530324325263004
  y: 0.017811554402223567
  z: 0.20422853198608543
  w: -0.9771975268789446
orientation_covariance: [1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0]
angular_velocity:
  x: -0.0005168463103473186
  y: 1.123920083045962e-05
  z: 0.00020210817456245422
angular_velocity_covariance: [1.2184696791468346e-07, 0.0, 0.0, 0.0, 1.218469679
1468346e-07, 0.0, 0.0, 0.0, 1.2184696791468346e-07]
linear_acceleration:
  x: 0.12748645
  y: 1.5984839500000012
  z: 10.1204628
linear_acceleration_covariance: [8.999999999999999e-08, 0.0, 0.0, 0.0, 8.9999999
```

QGC里解锁以后动手柄机器人螺旋桨没反应，不知道是哪里的问題

QGC里设置Vehicle Setup里把启动游戏手柄输入的钩钩上



有关检查手柄：

终端里：

`ls /dev/input/js*` 检查设备：

```
hit@hit-NUC8i5BEHS:~$ ls /dev/input/js*
/dev/input/js0
hit@hit-NUC8i5BEHS:~$
```

`sudo chmod 777 /dev/input/js0` 给权限

`roslaunch joy joy_node` 启动手柄ros节点：（如图则为检测到手柄信息）

```
hit@hit-NUC8i5BEHS:~$ roslaunch joy joy_node
[ WARN] [1727668512.195675768]: Couldn't set gain on joystick force feedback: Bad file descriptor
[ INFO] [1727668512.199980952]: Opened joystick: /dev/input/js0 (Logitech Logitech Dual Action). deadzone_: 0.050000.
```

检查话题：

```
hit@hit-NUC8i5BEHS:~$ rostopic list
/diagnostics
/joy
/joy/set_feedback
/rosout
/rosout_agg
```

```
hit@hit-NUC8i5BEHS:~$ rostopic echo /joy
header:
  seq: 1
  stamp:
    secs: 1727668622
    nsecs: 984244120
  frame_id: "/dev/input/js0"
axes: [0.0, 0.0, -0.08842818439006805, 0.0, 0.0, 0.0]
buttons: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
---
header:
  seq: 2
  stamp:
    secs: 1727668622
    nsecs: 992205294
  frame_id: "/dev/input/js0"
axes: [0.0, 0.0, -0.2620624303817749, 0.0, 0.0, 0.0]
buttons: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

正确输出包含axes和buttons的字段信息则说明手柄没有问题

5.安装DVL驱动与使用

[DVL A50](#)

创建好一个工作空间dvl_ws， dvl_ws/src文件夹下，克隆上面的功能包，返回工作空间catkin_make


```
hit@hit-NUC8i5BEHS: ~/dvl_ws
Scanning dependencies of target waterlinked_a50_ros_driver_generate_messages_cpp
Scanning dependencies of target waterlinked_a50_ros_driver_generate_messages_py
Scanning dependencies of target waterlinked_a50_ros_driver_generate_messages_lisp
[ 50%] Generating EusLisp code from waterlinked_a50_ros_driver/DVL.msg
[ 50%] Generating Javascript code from waterlinked_a50_ros_driver/DVLBeam.msg
[ 50%] Generating C++ code from waterlinked_a50_ros_driver/DVLBeam.msg
[ 50%] Generating Javascript code from waterlinked_a50_ros_driver/DVL.msg
[ 50%] Generating EusLisp code from waterlinked_a50_ros_driver/DVLBeam.msg
[ 50%] Generating C++ code from waterlinked_a50_ros_driver/DVL.msg
[ 58%] Generating Lisp code from waterlinked_a50_ros_driver/DVL.msg
[ 66%] Generating Python from MSG waterlinked_a50_ros_driver/DVL
[ 75%] Generating Lisp code from waterlinked_a50_ros_driver/DVLBeam.msg
[ 75%] Built target waterlinked_a50_ros_driver_generate_messages_nodejs
[ 83%] Generating EusLisp manifest code for waterlinked_a50_ros_driver
[ 91%] Generating Python from MSG waterlinked_a50_ros_driver/DVLBeam
[ 91%] Built target waterlinked_a50_ros_driver_generate_messages_lisp
[ 91%] Built target waterlinked_a50_ros_driver_generate_messages_cpp
[100%] Generating Python msg __init__.py for waterlinked_a50_ros_driver
[100%] Built target waterlinked_a50_ros_driver_generate_messages_py
[100%] Built target waterlinked_a50_ros_driver_generate_messages_eus
Scanning dependencies of target waterlinked_a50_ros_driver_generate_messages
[100%] Built target waterlinked_a50_ros_driver_generate_messages
hit@hit-NUC8i5BEHS:~/dvl_ws$
```

运行: `roslaunch waterlinked_a50_ros_driver publisher.py _ip:=192.168.194.95 _do_log_raw_data:=true`

报错: [ERROR] [1727669726.186719]: No route to host, DVL might be booting? [Errno 111]

Connection refused

待解决是因为ip的问题


```
roscore http:... x /home/hit/bl... x hit@hit-NUC... x hit@hit-NUC... x
hit@hit-NUC8i5BEHS:~$ rosruncat waterlinked_a50_ros_driver publisher.py _ip:=192.168.2.95 _do_log_raw_data:=true
[ERROR] [1727621101.225116]: No route to host, DVL might be booting? [Errno 113]
No route to host
[ERROR] [1727621104.297242]: No route to host, DVL might be booting? [Errno 113]
No route to host
^C[ERROR] [1727621107.369078]: No route to host, DVL might be booting? [Errno 113]
No route to host
[ERROR] [1727621110.441177]: No route to host, DVL might be booting? [Errno 113]
No route to host
[ERROR] [1727621113.513133]: No route to host, DVL might be booting? [Errno 113]
No route to host
```

小主机的ip设置为：192.168.194.90



运行 `roscuncat waterlinked_a50_ros_driver publisher.py _ip:=192.168.194.95 _do_log_raw_data:=true` 后报错: `TypeError: can only concatenate str (not "bytes") to str`

根据信息找到错误行: `raw_data = raw_data + rec` 试图将字节 (bytes) 类型与字符串 (str) 类型拼接, 这在Python3中是不允许的

更改为: `raw_data = raw_data + rec.decode('utf-8')` # 将bytes转换为str

```
30 def getData():
31     global oldJson, s
32     raw_data = ""
33
34     while not '\n' in raw_data:
35         try:
36             rec = s.recv(1) # Add timeout for that
37             if len(rec) == 0:
38                 rospy.logerr("Socket closed by the DVL, reopening")
39                 connect()
40                 continue
41             except socket.timeout as err:
42                 rospy.logerr("Lost connection with the DVL, reinitiating the
connection: {}".format(err))
43                 connect()
44                 continue
45             raw_data = raw_data + rec.decode('utf-8') # 将bytes转换为str
46         raw_data = oldJson + raw_data
47         oldJson = ""
48         raw_data = raw_data.split('\n')
49         oldJson = raw_data[1]
50         raw_data = raw_data[0]
51     return raw_data
52
```

修改后运行指令可以看到DVL数据输出:

```
hit@hit-NUC8i5BEHS: ~  
roscore http://hit-NUC8i5BEHS:11311/ x hit@hit-NUC8i5BEHS: ~ x  
valid":true},{ "id":1,"velocity":-0.0006256797350943089,"distance":0.059000000357  
62787,"rssi":-26.041929244995117,"nsd":-99.64151000976562,"beam_valid":true},{ "i  
d":2,"velocity":-0.00010893889702856541,"distance":0.05900000035762787,"rssi":-2  
5.879308700561523,"nsd":-90.01412963867188,"beam_valid":true},{ "id":3,"velocity"  
:-0.00013814447447657585,"distance":0.10620000213384628,"rssi":-29.3622436523437  
5,"nsd":-83.1578598022461,"beam_valid":true}], "velocity_valid":true,"status":0,"  
format":"json_v3","type":"velocity","time_of_validity":1647807417773063,"time_of  
_transmission":1647807417871883}  
[INFO] [1727786233.527156]: {"time":88.1509780883789,"vx":-0.0005012811161577702  
,"vy":0.0003144991642329842,"vz":-0.00006326884613372386,"fom":0.001275458722375  
3333,"covariance":[[0.000001555213998472027,-9.615430940357328e-8,-3.05153150748  
08314e-7],[-9.615430940357328e-8,9.749413720783195e-7,-1.0070733225120421e-7],[-  
3.0515315074808314e-7,-1.0070733225120421e-7,1.4300363204711175e-7]], "altitude":  
0.0504106730222702,"transducers":[{"id":0,"velocity":0.000534305814653635,"dista  
nce":0.05900000035762787,"rssi":-23.683565139770508,"nsd":-84.74211883544922,"be  
am_valid":true},{ "id":1,"velocity":-0.0006370211485773325,"distance":0.059000000  
35762787,"rssi":-26.043174743652344,"nsd":-100.71454620361328,"beam_valid":true}  
, {"id":2,"velocity":-0.00012196903117001057,"distance":0.05900000035762787,"rssi  
":-25.87882423400879,"nsd":-91.91093444824219,"beam_valid":true},{ "id":3,"veloci  
ty":-0.00016545108519494534,"distance":0.10620000213384628,"rssi":-29.3629512786  
86523,"nsd":-86.25208282470703,"beam_valid":true}], "velocity_valid":true,"status  
":0,"format":"json_v3","type":"velocity","time_of_validity":1647807417849174,"ti  
me_of_transmission":1647807417957905}
```

```
hit@hit-NUC8i5BEHS:~$ rostopic list  
/dvl/data  
/dvl/json_data  
/rosout  
/rosout_agg
```

6. 下载nomachine远程桌面软件

[nomachine](#) 下载

[安装教程](#) 按照教程将压缩包复制进/usr后操作


```

fyx@NUC11PAHi5:~$ su
密码:
root@NUC11PAHi5:/home/fyx# cd 下载
root@NUC11PAHi5:/home/fyx/下载# cp -r nomachine_8.14.2_1_x86_64.tar.gz /usr
root@NUC11PAHi5:/home/fyx/下载# cd
root@NUC11PAHi5:~# cd /usr
root@NUC11PAHi5:/usr# sudo tar zxvf nomachine_8.14.2_1_x86_64.tar.gz
NX/
NX/nxserver
NX/etc/
NX/etc/NX/
NX/etc/NX/server/
NX/etc/NX/server/localhost/
NX/etc/NX/server/localhost/server.cfg.sample
NX/etc/NX/server/packages/
NX/etc/NX/server/packages/nxnode.tar.gz
NX/etc/NX/server/packages/nxrunner.tar.gz
NX/etc/NX/server/packages/nxplayer.tar.gz
NX/etc/NX/server/packages/nxserver.tar.gz
README-NOMACHINE
root@NUC11PAHi5:/usr# sudo /usr/NX/nxserver --install
NX> 700 Starting installation at: Thu, 31 Oct 2024 17:23:39.
NX> 700 Using installation profile: Ubuntu.
NX> 700 Installation log is: /usr/NX/var/log/nxinstall.log.
NX> 700 Installing nxrunner version: 8.14.2.
NX> 700 Installing nxplayer version: 8.14.2.
NX> 700 Installing nxnode version: 8.14.2.
NX> 700 Installing nxserver version: 8.14.2.
NX> 700 Installation completed at: Thu, 31 Oct 2024 17:23:50.
NX> 700 NoMachine was configured to run the following services:
NX> 700 NX service on port: 4000
root@NUC11PAHi5:/usr#

```

小主机拔掉HDMI线后系统会自动关闭显示输出，因此远程桌面会黑屏，解决办法：

（因为已经装进密封舱连好线了，因此尝试以下不需要联网的方法）

1) 在另一台联网的电脑上下载.deb包

```

1 sudo apt update
2 apt download xserver-xorg-video-dummy

```

```

hit@hit-NUC8i5BEHS:~$ apt download xserver-xorg-video-dummy
获取:1 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal/main amd64 xserver-xorg
-video-dummy amd64 1:0.3.8-1build3 [8,884 B]
已下载 8,884 B, 耗时 0秒 (26.8 kB/s)

```

这会在主目录下载.deb安装包，用U盘将此文件拷入小主机



xserver-
xorg-video-
dummy_1...

2) 小主机中,

```
sudo dpkg -i xserver-xorg-video-dummy_*.deb
```

3) 在文件夹 /usr/share/X11/xorg.conf.d中, 创建xorg.conf文件, 写入以下内容:

```
1  Section "Device"
2      Identifier "Configured Video Device"
3      Driver "dummy"
4      VideoRam 256000
5  EndSection
6
7  Section "Monitor"
8      Identifier "Configured Monitor"
9      HorizSync 28.0-80.0
10     VertRefresh 48.0-75.0
11     Modeline "1920x1080_60.00" 172.80 1920 2040 2248 2576 1080 1081 1084 1
118 -HSync +Vsync
12 EndSection
13
14 Section "Screen"
15     Identifier "Default Screen"
16     Monitor "Configured Monitor"
17     Device "Configured Video Device"
18     DefaultDepth 24
19     SubSection "Display"
20         Depth 24
21         Modes "1920x1080_60.00"
22     EndSubSection
23 EndSection
```

4) 保存, 重启系统

此时拔掉HDMI线, 连接nomachine就可以控制了

7.动捕配置及使用

需要下载相关的功能包：

3. VRPN 客户端的下载及网络配置，使用虚拟机运行（如图 9-11）

- `cd ~/catkin_ws/src`
- `git clone https://github.com/clearpathrobotics/vrpn_client_ros.git`
- `sudo apt-get install ros-kinetic-vrpn`

使用说明  [Seeker与ROS的通信.pdf](#)